

■ Plot

`Plot[f, {x, xmin, xmax}]` generates a plot of f as a function of x from $xmin$ to $xmax$.

`Plot[{f1, f2, ...}, {x, xmin, xmax}]` plots several functions f_i .

`Plot` evaluates its arguments in a non-standard way (see page ??). ■ The following options can be given:

<code>AspectRatio</code>	<code>1/GoldenRatio</code>	ratio of height to width
<code>Axes</code>	<code>Automatic</code>	whether to include axes
<code>AxesLabel</code>	<code>None</code>	axes labels
<code>DisplayFunction</code>	<code>\$DisplayFunction</code>	function for generating output
<code>Framed</code>	<code>False</code>	whether to draw a frame
<code>MaxBend</code>	<code>10.</code>	maximum bend between segments
<code>PlotColor</code>	<code>True</code>	whether to plot in color
<code>PlotDivision</code>	<code>20.</code>	maximum subdivision factor in sampling
<code>PlotLabel</code>	<code>None</code>	a label for the plot
<code>PlotPoints</code>	<code>25</code>	initial number of sample points
<code>PlotRange</code>	<code>Automatic</code>	range of z values to include
<code>PlotStyle</code>	<code>Automatic</code>	graphics primitives to specify the style for each curve
<code>Ticks</code>	<code>Automatic</code>	tick marks

■ `Plot` initially evaluates f at a number of equally-spaced sample points specified by `PlotPoints`. Then it uses an adaptive algorithm to choose additional sample points, attempting to produce a curve in which the bend between successive segments is less than `MaxBend`. It subdivides a given interval by a factor of at most `PlotDivision`. ■ You should realize that with the finite number of sample points used, it is possible for `Plot` to miss features in your function. To check your results, you should increase the setting for `PlotPoints`. ■ `Plot` returns a `Graphics` object. ■ See page 107. ■ See also: `ListPlot`, `Graphics`.